

Raspberry Pi-Serial Port Method

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Experimental preparation

- 1. Raspberry Pi motherboard
- 2. 8-channel line patrol module
- 3. Several Dupont cables

The Raspberry Pi board needs to download the serial communication source code provided in the document, and the Raspberry Pi system needs to open the serial communication interface through raspi-config

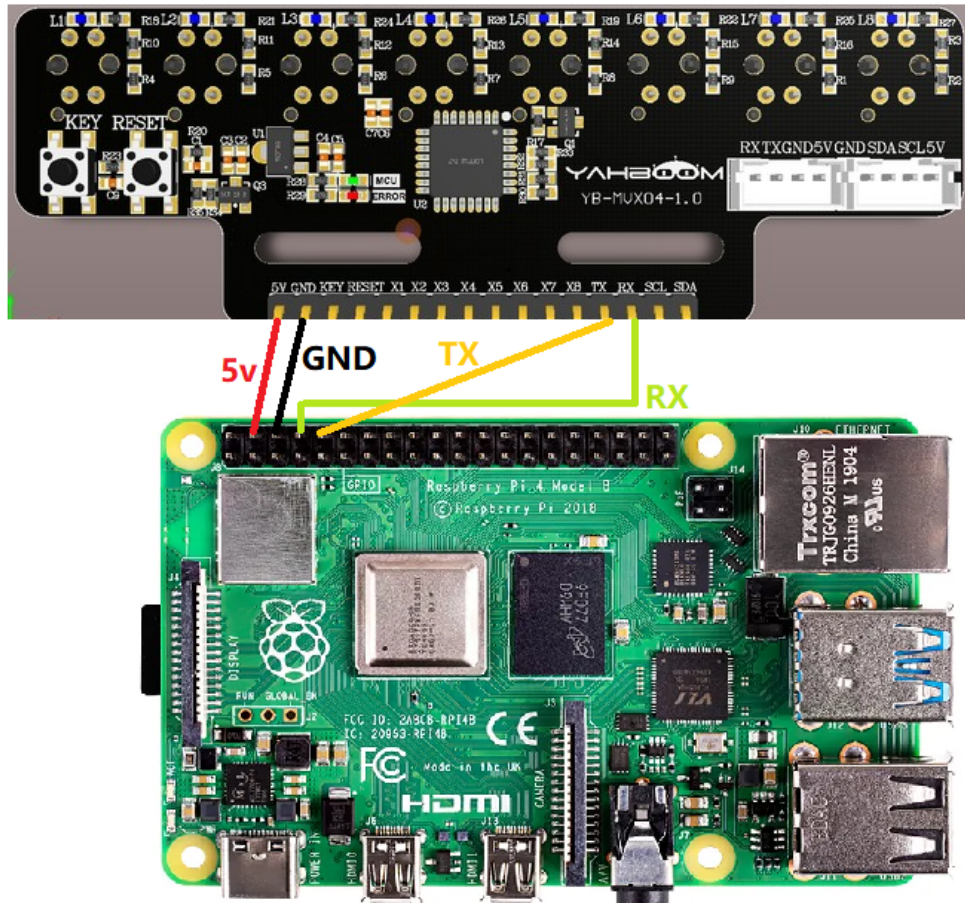
Experimental purpose

The content of this experiment is mainly to use the Raspberry Pi main control to receive the data of the 8-channel line patrol module through the serial port.

Experimental wiring

Raspberry Pi	8-channel line patrol module
TX	RX
RX	TX
5v	5v
GND	GND

As shown in the figure:

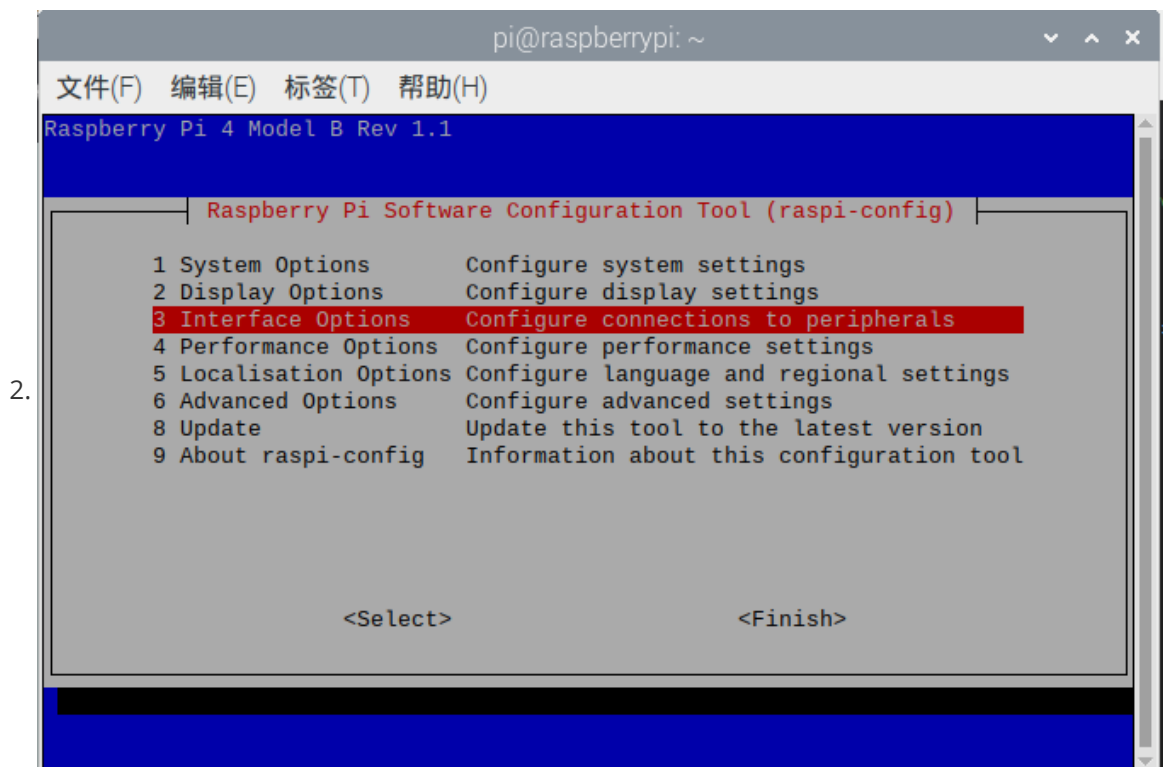


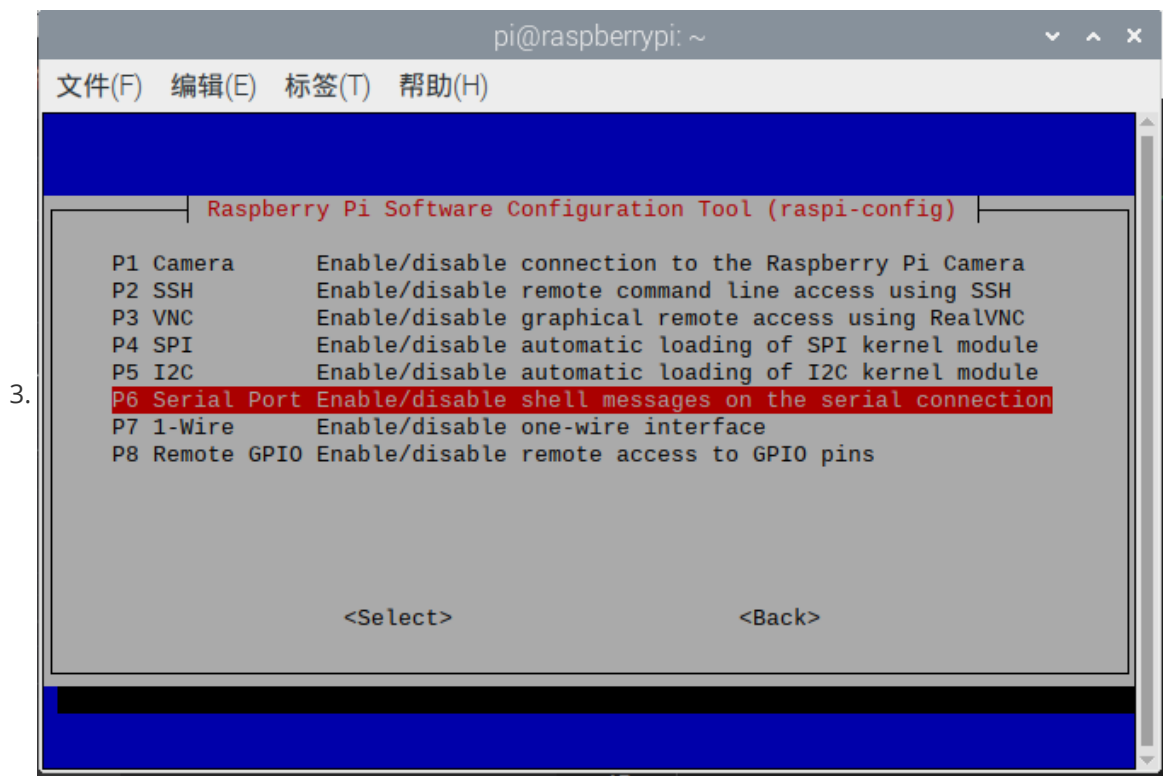
Open the Raspberry Pi hard serial port (Raspberry Pi 5 does not require this step)

- Configure the Raspberry Pi serial port first, because the Raspberry Pi's hard serial port is used for Bluetooth, and the mini serial port is unstable. This experiment uses the hard serial port. Raspberry Pi pin diagram

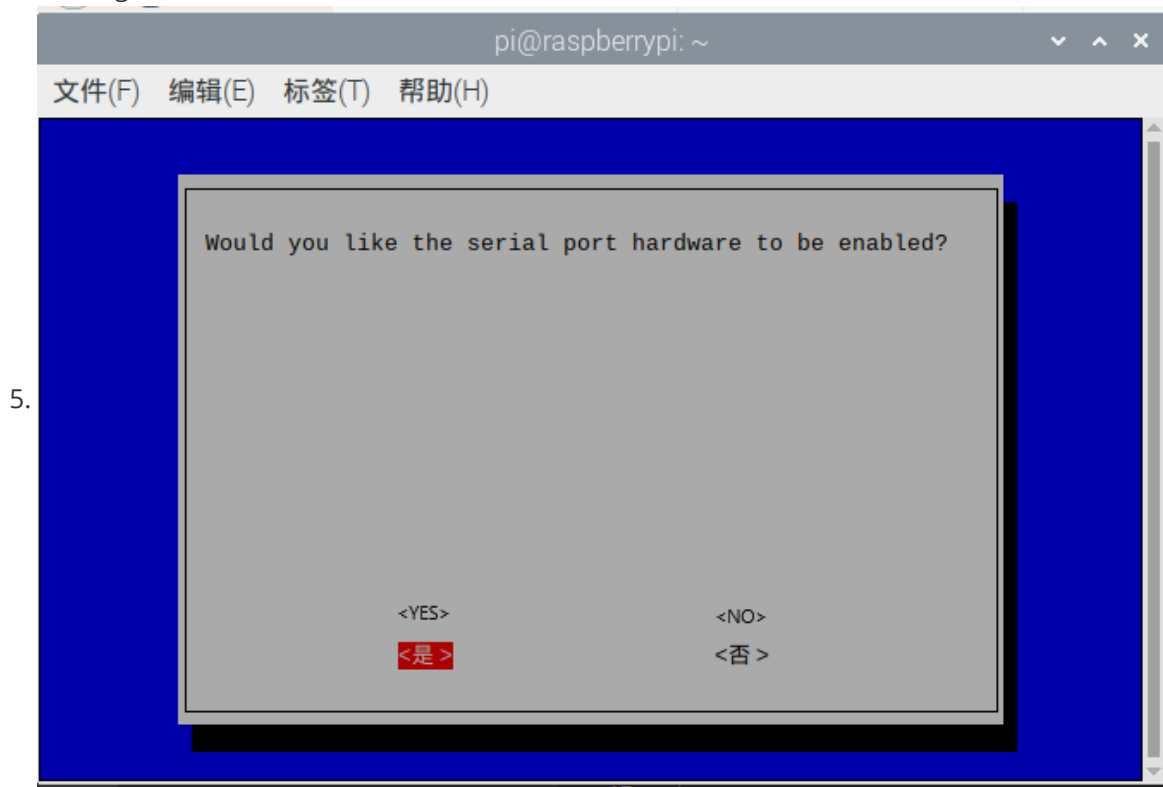
wiringPi	BCM	Funtion	Physical pin		Funtion	BCM	wiringPi
		3.3V	1	2	5V		
8	2	SDA.1	3	4	5V		
9	3	SCL.1	5	6	GND		
7	4	GPIO.7	7	8	TXD	14	15
		GND	9	10	RXD	15	16
0	17	GPIO.0	11	12	GPIO.1	18	1
2	27	GPIO.2	13	14	GND		
3	22	GPIO.3	15	16	GPIO.4	23	4
		3.3V	17	18	GPIO.5	24	5
12	10	MOSI	19	20	GND		
13	9	MISO	21	22	GPIO.6	25	6
14	11	SCLK	23	24	CE0	8	10
		GND	25	26	CE1	7	11
30	0	SDA.0	27	28	SCL.0	1	31
21	5	GPIO.21	29	30	GND		
22	6	GPIO.22	31	32	GPIO.26	12	26
23	13	GPIO.23	33	34	GND		
24	19	GPIO.24	35	36	GPIO.27	16	27
25	26	GPIO.25	37	38	GPIO.28	20	28
		GND	39	40	GPIO.29	21	29

1. First perform the following operations to map the serial port Enter `sudo raspi-config` in the terminal





4.  image-2022120500010



6. Set the hardware serial port to GPIO serial port, edit /boot/config.txt with root privileges.

The command is:

```
sudo nano /boot/config.txt
```

After opening the file, add two lines at the end

```
dtoverlay=miniuart-bt  
force_turbo=1
```

```
GNU nano 3.2 /boot/config.txt

#dtoverlay=lirc-rpi

# Additional overlays and parameters are documented /boot/overlays/README

# Enable audio (loads snd_bcm2835)
dtparam=audio=on
start_x=1
gpu_mem=128

dtoverlay=pi3-miniuart-bt
force_turbo=1
```

Save: Ctrl+O, exit: Ctrl+X.

1. Save and exit, then restart the Raspberry Pi and you can see that the serial ports have been swapped.

```
pi@raspberrypi: ~
文件(F) 编辑(E) 标签(T) 帮助(H)

pi@raspberrypi:~ $ ls -l /dev/serial*
lrwxrwxrwx 1 root root 7 10月 16 12:06 /dev/serial0 -> ttyAMA0
lrwxrwxrwx 1 root root 5 10月 16 12:06 /dev/serial1 -> ttyS0
pi@raspberrypi:~ $
```

- 3.
4. Reference Links: <https://blog.iyatt.com/?p=1817>

Experimental steps and phenomena

1. After connecting the wires, run the script

```
python3 UASRT.py
```